

Class 10: Structural Bioinformatics

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB)

PDB statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats from <https://www.rcsb.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	217,375
2	1	0	14,283
3	0	0	16,277
4	3	1	5,004
5	0	0	244
6	0	4	22

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

The print out above `pdb$X.ray` is “character” not “numeric” Therefore I can’t do math with it. We need to fix this...

Two functions that can help here are `sub()` and `as.numeric()`

```
# We want to get rid (or sub out) commas
x <- pdb$X.ray
tmp <- sub( ",", "", x)
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma <- function(x) {
  tmp <- sub( ",", "", x)
  sum(as.numeric(tmp))
}
```

```
rm.comma(pdb$EM)
```

```
[1] 33880
```

We could also use a different import function for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file).

```
library(readr)

pdb <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
-- Column specification -----
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
pdb
```

```
# A tibble: 6 x 9
  `Molecular Type`    `X-ray`    EM    NMR Integrative `Multiple methods` Neutron
  <chr>              <dbl> <dbl> <dbl>      <dbl>          <dbl>  <dbl>
1 Protein (only)      180758 23111 12813        348            229    84
2 Protein/Oligosacch~  10488  3741   34         8             11     1
3 Protein/NA          9205  6751  287        26             8     0
4 Nucleic acid (only)  3154   250 1578         3            15     3
5 Other               178    27   35         4             0     0
6 Oligosaccharide (o~  11     0    6         0             1     0
# i 2 more variables: Other <dbl>, Total <dbl>
```

```
n.tot <- pdb$Total
sum(pdb$`X-ray`) / n.tot
```

```
[1] 0.9375227 14.2682910 12.5203662 40.7262190 835.2213115
[6] 9263.3636364
```

```
n.tot <- sum(pdb$Total)
n.xray <- sum(pdb$`X-ray`)
n.em <- sum(pdb$EM)
```

```
n.xray/n.tot*100
```

```
[1] 80.48577
```

```
n.em/n.tot*100
```

```
[1] 13.38046
```

Q. How many total protein structures are there in the dataset?

```
pdb$Total[1]
```

```
[1] 217375
```

The total number of protein sequences in UniProt is 202,556,314

```
217375/202556314 *100
```

```
[1] 0.1073158
```

Key-point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography).

Visualizing PDB data with Mol-star

Main stand alone web version with all features is at <https://molstar.org/viewer/>

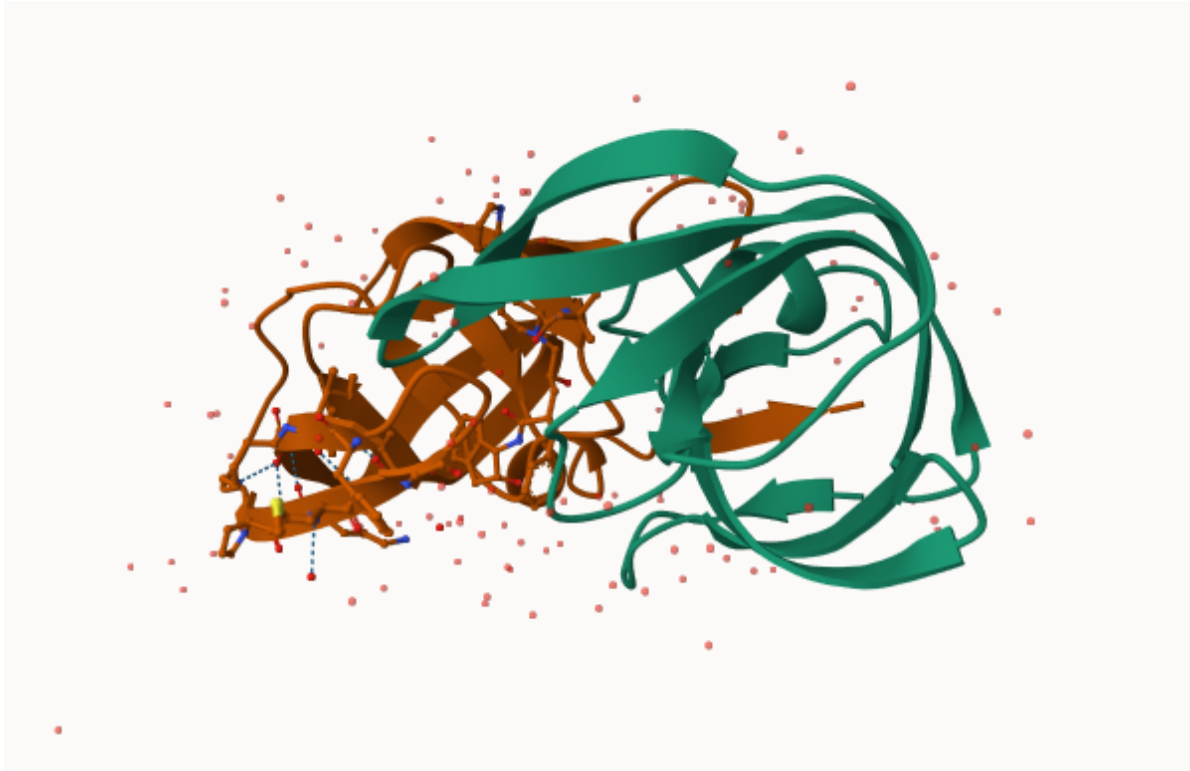


Figure 1: The HIV-protease enzyme is a homodimer with two chains

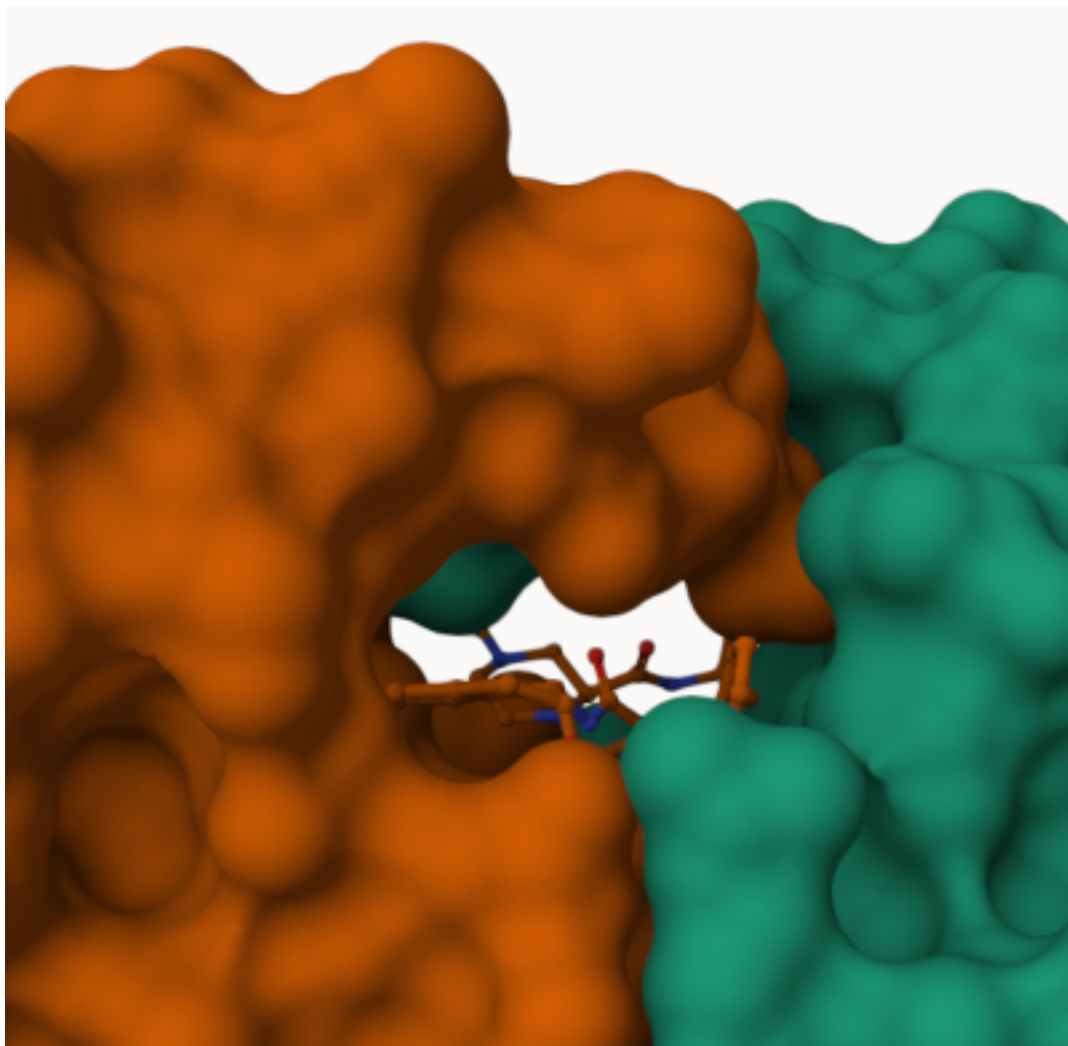


Figure 2: Surface display showing the binding cleft site for the inhibitor (drug) molecule

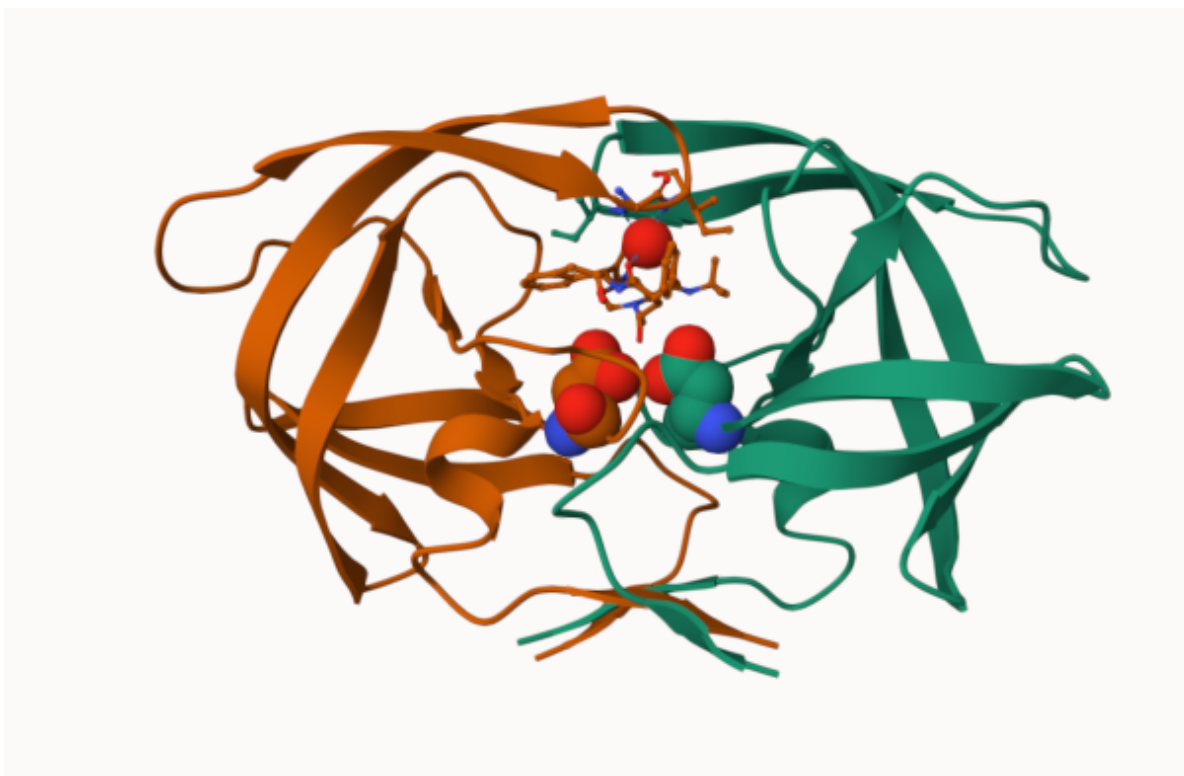


Figure 3: Spacefill display of catalytic ASP25 amino acids and key binding site water molecule

Getting started with the Bio3D package

Bio3D is an R package from CRAN for structural bioinformatics

```
library(bio3d)

pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)

Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

`attributes(pdb)`

\$names

[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"

\$class

[1] "pdb" "sse"

`head(pdb$atom)`

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

There are lots of functions that can work with these `pdb` objects

```
head(pdbseq(pdb))
```

```
  1    2    3    4    5    6  
"P" "Q" "I" "T" "L" "W"
```

We can have a quick interactive view of any of these `pdb` objects:

```
library(bio3dview)  
  
#view.pdb(pdb)
```

Lets try a custom view

Q. Create a custom view of HIV-Pr highlighting the active site (`resno=25`), the two chains (in your choice of colors), and the ligand all on a custom color background?

```
library(NGLViewerR)  
# active.site <- atom.select(pdb, rsno=25)  
  
# view.pdb(pdb,  
#         cols = c("red","blue"),  
#         highlight = active.site,  
#         highlight.style = "spacefill",  
#         backgroundColor = "pink") %>%  
#   setRock()
```

Predicting functional motions of a single structure

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, `rm.alt=TRUE`

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
TDELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

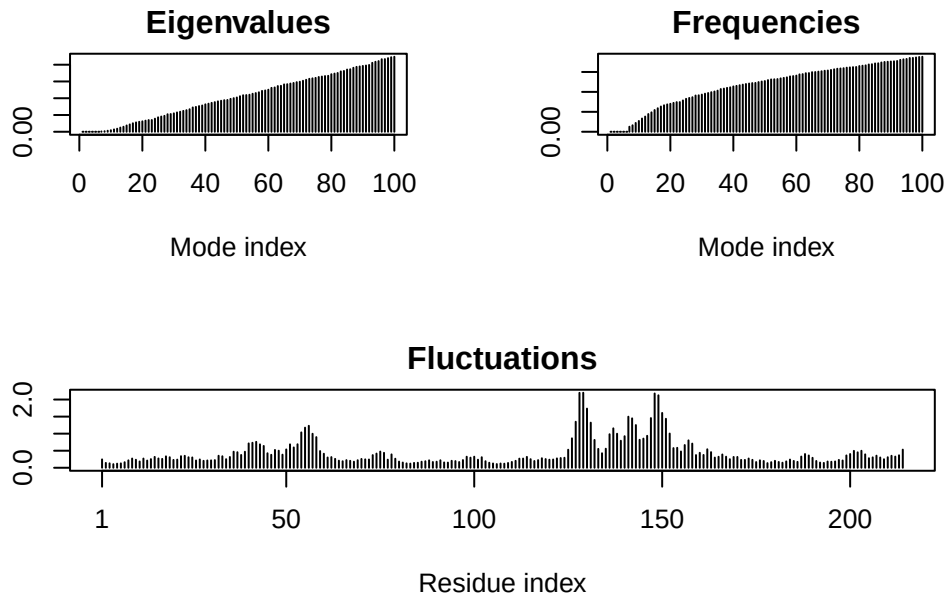
```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian... Done in 0.02 seconds.
```

```
Diagonalizing Hessian... Done in 0.27 seconds.
```

```
plot(m)
```



```
mktrj(m, file="adk_m7.pdb")
# view.nma(m, pdb=adk)
```

Comparative structure analysis of Adenylate Kinase

```
## Install packages in the R console NOT your Rmd/Quarto file

#install.packages("bio3d")
#install.packages("NGLVieweR")

#install.packages("remotes")
#remotes::install_github("bioboot/bio3dview")

#install.packages("BiocManager")
#BiocManager::install("msa")
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

The package "msa" is only found on BioConductor

Q11. Which of the above packages is not found on BioConductor or CRAN?

"bio3dview" is not found on BioConductor or CRAN.

Q12. True or False? Functions from the **pak** package can be used to install packages from GitHub and BitBucket?

True

Search and retrieve ADK structures

```
library(bio3d)
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
#aa
```

Q13. How many amino acids are in this sequence?

There are 214 amino acids.

Aligning and superposing structures

```
hits <- NULL
hits$ pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A',
                 '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A',
                 '6HAP_A', '6HAM_A', '4K46_A', '3GMT_A',
                 '4PZL_A')
```

```
# Download related PDB files
files <- get.pdb(hits$ pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

	0%
=====	8%
=====	15%
=====	23%



```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
```

```
.... PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
...
```

Extracting sequences

```
pdb/seq: 1 name: pdbs/split_chain/1AKE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2 name: pdbs/split_chain/6S36_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3 name: pdbs/split_chain/6RZE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4 name: pdbs/split_chain/3HPR_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5 name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6 name: pdbs/split_chain/5EJE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7 name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8 name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9 name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10 name: pdbs/split_chain/6HAM_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11 name: pdbs/split_chain/4K46_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12 name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13 name: pdbs/split_chain/4PZL_A.pdb
```

Annotating PDB structures

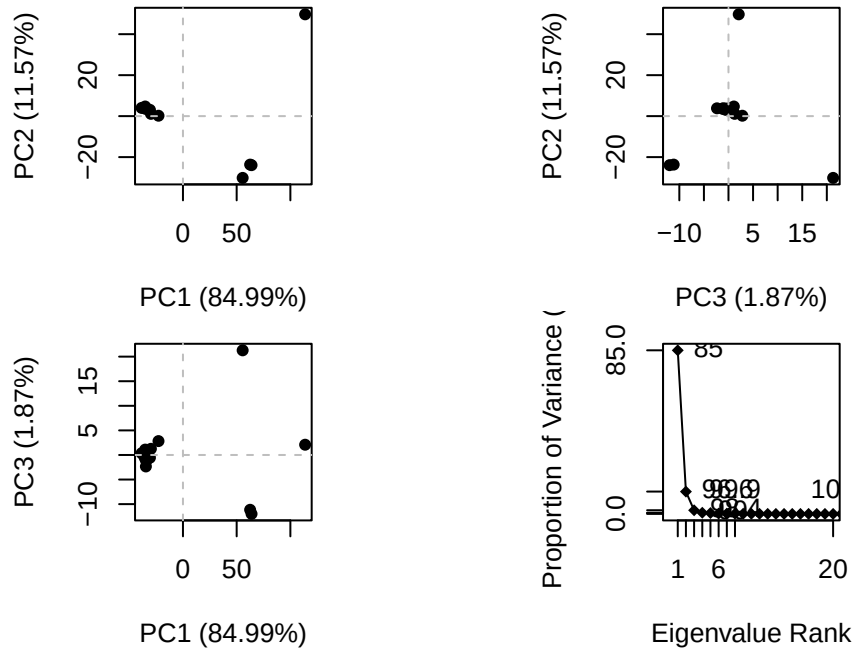
```
# Vector containing PDB database codes
ids <- basename.pdb(pdb$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

Principal Component Analysis

```
# Perform PCA
pc.xray <- pca(pdbbs)
plot(pc.xray)
```

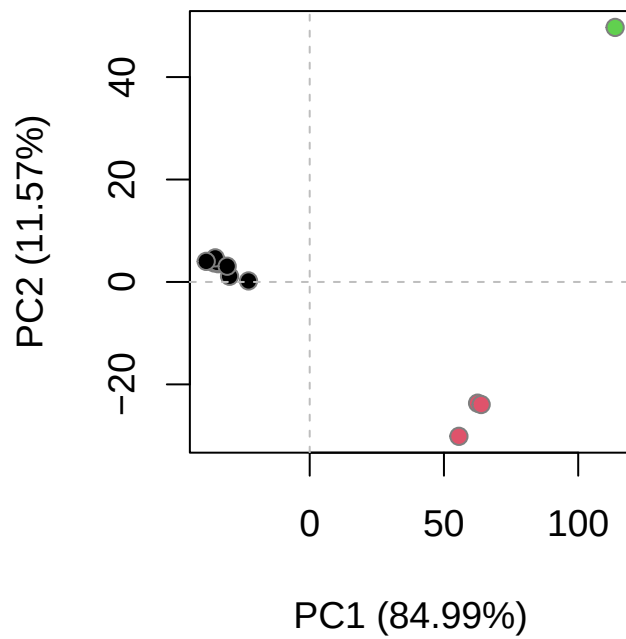


```
# Calculate RMSD
rd <- rmsd(pdbbs)
```

Warning in rmsd(pdbbs): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



```
# Visualize first principal component
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                 PC2=pc.xray$z[,2],
                 col=as.factor(grps.rd),
                 ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```